AUTHBC — Running Results Summary

Co-optimizing encoding × authentication placement × signature scheme × batching for blockchain-grade UAV telemetry over 802.11

Mohamed A. Farouk
(living document — regenerated as each step completes)

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Abstract

This document tracks every measured result in the AUTHBC project and is updated at each step. All numbers trace to a frozen CSV in `results/` and are guarded by the reproduction gate (`make verify-frozen`). The headline is established: the co-optimized configuration cuts on-air authentication bytes by **96.4 %** versus the inline-CBOR baseline while holding verifiability $V \geq 0.95$ at loss $p = 0.05$ — criterion met. The hardware campaign has confirmed the one predicted anomaly (the signature-scheme ordering flips on ARM) and has shown the energy model's assumed CPU power to be 4.1× too high.

1 The problem

Telemetry records are tiny (45–190 B) but a signature is 64–96 B, so authentication overhead rivals the payload. Four knobs — record *encoding*, authentication *placement* (A inline, B self-batch, C relay-aggregate, D block), signature *scheme*, and *batch* size b — are coupled and must be co-optimized. The falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by $\geq 40\,\%$ versus inline-CBOR at $V \geq 0.95$, $p = 0.05$.

2 Software layer (frozen, reproducible)

2.1 E1 — overhead dominance (T1)

$\varphi = g/(s + g)$ is the fraction of on-air bytes that is pure signature ($g = 64\,\text{B}$).

encoding	mean record s	φ
JSON	191.1 B	25.1 %
CBOR	66.25 B	49.1 %
MessagePack	65.16 B	49.6 %
delta	**45.00 B**	**58.7 %**

Reading. The better the compression, the *worse* the relative overhead: delta is the best encoder and leaves 59 % of the frame as signature. Compression alone cannot solve this — the motivation for the other three knobs.

2.2 E2 — batching cure (T2)

Folding b records under one signature gives $s+(g_a+H_f)/b$ per record; packing to the MTU bottoms out at $A = M/(M - H_f - g_a) = 1.0745$ at $M = 1500$. Measured: delta reaches $b_{\max} = 31$ and attains $A = 1.0745$ exactly; other encodings within 5 % (integer-b slack). Inline placement *never* amortizes its per-record signature — only the header shrinks.

2.3 E3 — loss frontier (T3)

Self-batch frames verify independently, so $V_B = 1 - p$, *flat in b*. A block signature spanning n frames needs all fragments: $V_D = (1-p)^n$. Measured at $p = 0.10$, $b = 40$: $V_B = 0.90$ versus $V_D = 0.81 = 0.9^2$. This is the wall that stops over-aggregation.

2.4 E4 — scheme crossover (T4)

BLS aggregates but its pairing verify is 10.7× costlier than Ed25519. Which wins depends on the powers only through $\kappa = P_r/P_c$, so the break-even $\kappa^\star$ is found without measuring watts. **Result: Ed25519 wins all 80 grid points**, $\min \kappa^\star = 3.21$. At the accepted 96 B, BLS carries *more* bytes than Ed25519's 64 B on own traffic ($\kappa^\star = \infty$); its aggregation pays off only for cross-signer relay traffic.

2.5 E5 — the headline (T5) PASS

config	enc	scheme	placement	auth B/rec	V
optimized	delta	ECDSA	B, b=28	**3.71**	0.95
A+CBOR (baseline)	cbor	Ed25519	A inline	104.0	0.95
A+JSON (naive)	json	Ed25519	A inline	104.0	0.95
D-over-agg	cbor	Ed25519	D, b=40	3.60	**0.9025** FAIL

$$\text{cut} = \frac{104 - 3.71}{104} = \mathbf{96.4}\,\% \ \geq 40\,\% \quad \Rightarrow \quad \text{PASS}$$

Hand-checked: $45.0 + 104/28 = 48.71$ B/record. **The last row is the co-design argument in miniature**: over-aggregation is byte-competitive (3.60 B) yet *fails* $V \geq 0.95$ because $0.9025 = (1-p)^2$. The byte-optimal answer is not the correct answer.

2.6 NS-3 channel validation

N	unicast gap	broadcast gap
5	5.1 %	4.4 %
20	5.2 %	37 %
50	**1.8 %**	1735 %

Unicast validates the Bianchi DCF model to ±1.8–5.3 %. The broadcast column is discussed in §4 — part real capture effect, part an artefact of node placement.

3 Hardware layer (Raspberry Pi 4B, measured)

3.1 Cross-platform determinism

All **12/12** encoded-size metrics are byte-identical between x86 and ARM, as deterministic integer-only encoders require.

3.2 Timings and cycle normalisation

op	x86	RPi4	wall ratio	**cycles ARM/x86**
ed25519 sign	29.1 µs	88.1	3.03×	**1.16**×
ed25519 verify	95.0	259.5	2.73×	**1.05**×
ecdsa sign	26.2	135.5	5.17×	1.98×
ecdsa verify	78.5	326.9	4.16×	1.60×
bls verify	1016.5	8611.8	8.47×	3.25×

Wall-clock ratio conflates clock speed with per-cycle efficiency. The clock floor is 4.7/1.8 = 2.61×. Normalising shows Ed25519 needs only 1.05× the cycles on ARM (equally optimised on both ISAs) while BLS needs 3.25× (heavily x86-tuned). **Consequence: the docs/04 §1 "5–15×" anchor is unreachable** for an equally-optimised primitive and must be restated per-cycle.

3.3 F6 — the scheme ordering flips on ARM

platform	ECDSA verify	Ed25519 verify	winner
x86	78.5 µs	95.0 µs	ECDSA
RPi4	**326.9**	**259.5**	**Ed25519**

Three candidate causes were separated:

- **Library — ruled out.** Byte-identical versions on both (`cryptography` 45.0.7).
- **Our wrapper — real but not the cause.** Our ECDSA path does ASN.1 DER conversion in Python every op (Ed25519 does not): 1.05 µs on x86, **12.58 µs on ARM** (∼ 12× worse). Flipping the result would have required 64×. Core-only on ARM: ECDSA 312.4 versus Ed25519 263.3 — **Ed25519 still wins**.
- **Hardware — the operative cause**, but see §4: the *specific* microarchitectural mechanism is **not** established.

Consequence: on the real deployment platform the optimum selects **Ed25519** (the D2 architectural default). Both are 64 B, so the 96.4 % headline is **unchanged**.

3.4 Measured per-op energy (RPi4, 60 s windows, 5 reps)

op	energy/op (µJ)	95 % CI	ΔP (W)
ed25519 sign	**70.64**	—	0.773
ed25519 verify	**187.47**	—	0.717
ecdsa sign	87.10	—	0.634
ecdsa verify	201.04	—	0.613
bls sign	1397.73	—	0.513
bls verify	4555.00	—	0.529
delta encode	36.92	—	0.726
cbor encode	40.43	—	0.753

Ed25519 is cheaper than ECDSA in energy as well as time on ARM (70.6 vs 87.3 µJ sign; 186.8 vs 203.5 verify), consistent with the timing flip. Independent cross-check against $\Delta P \times t_{op}$ (micro suite) agrees within 1–4.4 % for seven of eight ops; all eight ops now agree within **3.7 %** (BLS to 0.0 % and −0.1 %). The earlier 14.4 % `bls:verify` discrepancy was traced to warmup running *inside* the sync window and was fixed at source and **re-measured** (superseded data retained under `results/hw/energy/superseded/`). Measured p_{cpu} median **0.634 W** (range 0.513–0.773).

3.5 Reproducibility and energy

Two boards, independent runs: worst-case divergence **0.5 %** across all 16 timed ops. The energy chain validated end-to-end (12 windows → 12 segments) with two independent cross-checks (loop throughput versus micro suite within 5–6 %; single-core ΔP 0.728 W versus 4-core stress 0.690 W/core). **Measured $p_{cpu} \approx 0.73$ W against the nominal 3.0 W used in E5** — 4.1× **too high.** The auth-byte headline is power-free and unaffected.

4 Open items and inconsistencies

item	status
F6 mechanism	**Claim retracted.** Masking ADX/BMI2 on x86 via `OPENSSL_ia32cap` did *not* slow ECDSA (61.1 → 56.8 µs; ECDSA still won), so the ADX explanation is **unsupported**. What remains established: the ordering is ISA-dependent and Ed25519 is markedly more portable (1.05× versus 1.60× cycles). The specific mechanism is an open question.
NS-3 broadcast	Nodes sit on a *line* at 1 cm spacing under a log-distance model ($n = 3$, $d_0 = 1\,\text{m}$), giving a **50.7 dB** near–far power spread. Capture needs only a few dB, so the geometry manufactures the very effect blamed for the gap. Needs an equal-power channel to test the model on its own assumptions.
κ plausible band	E4 assumes $\kappa = P_r/P_c \lesssim 0.5$ from "radio below CPU power". With p_{cpu} measured at 0.73 W (not 3.0 W), κ could approach or exceed 1. The E4 verdict is unchanged ($\min \kappa^\star = 3.21$) but the stated band must be re-derived from measured powers.
p_{radio}	**pending** — needs the 2-node link run; both boards now healthy.
E4/E5 re-run	**pending** — deliberately deferred until both powers are measured, so the energy column stays internally consistent rather than half-measured.
5–15× anchor	Needs restating per-cycle (decision required).

Provenance

Every number above traces to a frozen CSV under `results/`; derived artefacts are regenerated byte-identically by `make verify-frozen`, which fails loudly on any drift. Full audit: `docs/audits/p7.md`; decisions and their limitations: `docs/DECISIONS.md`.